

CP-322-A Assignment 1 Q1

Jordan Tauro, Clinton Osawe, Visula Jayasinghe, Philip
Marian

K-Nearest neighbours is an extremely simple yet effective way to classify data in machine learning. In this example, we found that the accuracy, when predicting the 15% validation set, peaked at 80% when k was equal to 3 and 5. Since the accuracy was tied, 3 was chosen as the best k number. When testing with the 15% test set, the accuracy of $k = 3$ lowered to 78%. When testing with test_inputs.csv and test_labels.csv, the accuracy fell even lower to 75.45%. Lastly, when calculating the accuracy using a 10-fold cross-validation, the accuracy slightly jumped to 78.30% (while still using $k = 3$).

